

Title: Design Document: PS13 - Communication Networks

Group ID: 61

Group Members

SL. No.	Name	BITS ID
1.	Ravinder Kumar	2025aa05445
2.	Kandarp Prakash	2025aa05547
3.	Vaishali Thankur	2025aa05635
4.	Vinay	2025aa05380
5.	Dhanashreenydh	2025aa05243

1. Executive Summary

This project implements and compares two distinct routing algorithms Dijkstra's Algorithm and Ant Colony Optimization (ACO) to solve the minimum-latency path problem in a simulated computer communication network. The network is modeled as an undirected, weighted graph where nodes represent routers and edges represent communication links with specific transmission latencies.

2. Data Structures & Methodology

To ensure optimal execution, the following data structures were utilized:

- **Adjacency List (Dictionary):** The network graph is stored as a dictionary where keys are nodes and values are lists of tuples (neighbor, latency). This allows for efficient $O(1)$ lookup of neighboring routers.
- **Min-Priority Queue (heapq):** Used in Dijkstra's algorithm to continually fetch the next unvisited node with the lowest cumulative latency, ensuring an efficient time complexity of $O((V+E) \log V)$.
- **Edge-Mapped Dictionaries:** Pheromone levels (τ) and Heuristic desirability (η) are stored in dictionaries keyed by edge tuples (u, v) . This allows $O(1)$ read/write access when ants make probabilistic routing decisions or deposit pheromones.

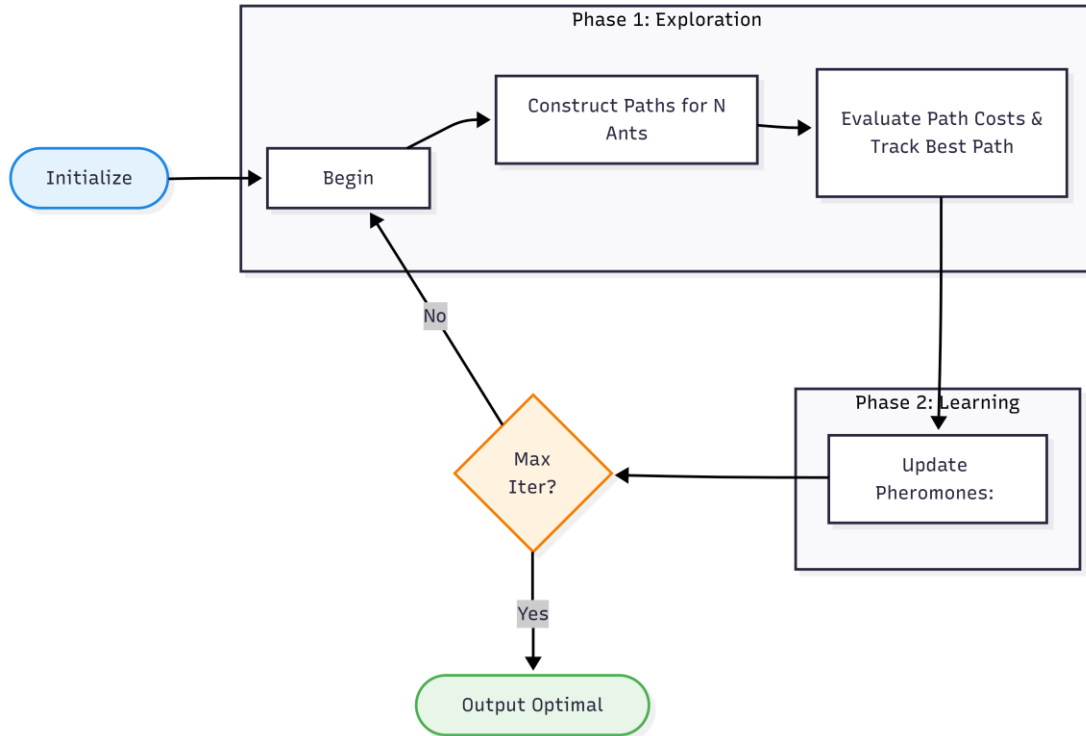


Figure.1 Ant path construction

3. Problem Formulation and PEAS Framework

The problem requires an intelligent routing agent to dynamically identify a minimum-latency path between a source and destination router under high traffic conditions.

- **Performance Measure:** Minimizing the total transmission latency (path cost) and minimizing the number of iterations required for the algorithm to converge on the optimal route.
- **Environment:** A fully observable, deterministic, static, and discrete graph environment.
- **Actuators:** The mechanism by which artificial ants traverse from a current router to an adjacent unvisited router, and the mathematical deposit/evaporation of virtual pheromones on the utilized communication links.
- **Sensors:** The agent's ability to sense available neighboring nodes, read the latency (heuristic desirability, $\eta = 1/\text{weight}$) of connecting edges, and detect the current pheromone concentration (τ) on those paths.

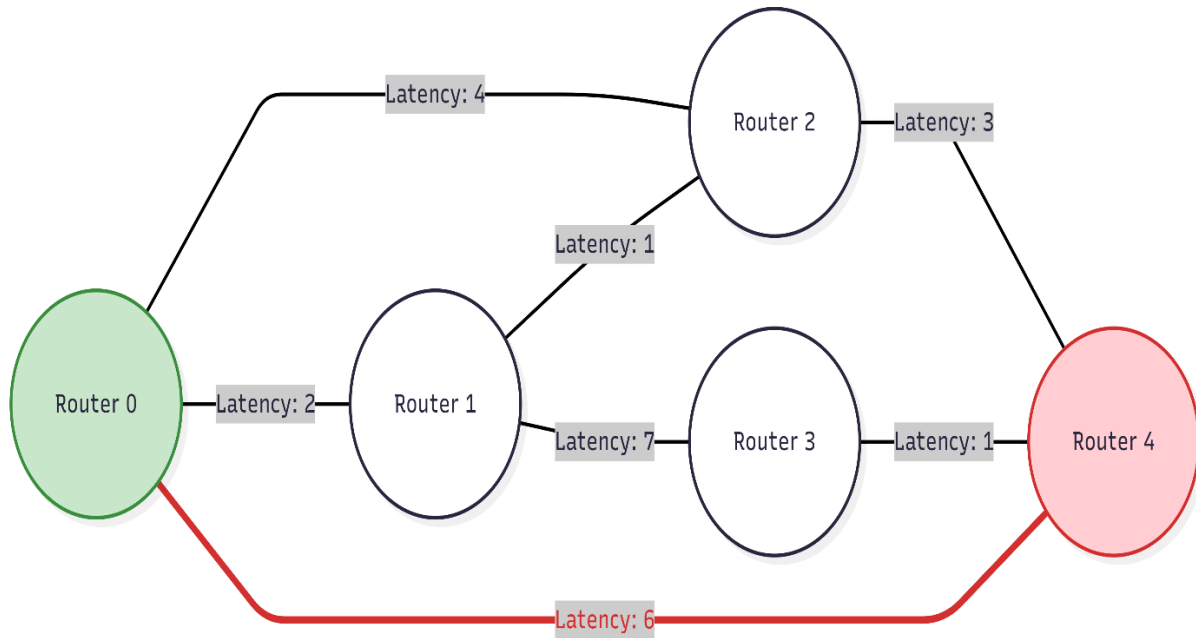


Figure.2 Router latency flow network graph

4. Alternative Problem Modeling & Performance Implications

Alternative Model: Reinforcement Learning (Q-Learning via MDP)

Instead of ACO (a population-based meta-heuristic), this problem can be modeled as a Markov Decision Process (MDP) solved via Q-Learning.

- **Modeling:** Each router acts as a "State". Moving across a link to a neighboring router is an "Action". The "Reward" is the inverse of the transmission latency. The agent updates a Q-Table mapping every state-action pair to a maximum expected future reward.
- **Performance Implications:**
 - * *Training Overhead:* Q-learning requires significant initial exploration time (many episodes) to adequately populate the Q-table, making it slower to converge initially than ACO for a single isolated source-destination query.
 - *Execution Speed:* Once the Q-table has converged, finding the shortest path from *any* source to *any* destination in the network is extremely fast ($O(1)$ table lookup per step). ACO, conversely, must be entirely rerun from scratch if the source or destination changes.

5. Performance Analysis

Dijkstra's deterministic algorithm found the absolute baseline minimum latency path of **6** (via path 0 -> 4). Under ACO testing, Scenario 1 achieved a cost of **6** (via path 0 -> 4) and Scenario 2 also achieved a cost of **6** (via path 0 -> 1 -> 2 -> 4).

Both ACO scenarios successfully matched the deterministic optimal path cost, though they discovered different physical routes that resulted in the exact same total minimum latency.

Convergence Speed: Both scenarios converged at the exact same speed, discovering their optimal paths immediately at iteration **1**.

- **Effect of Parameter Settings α , β , ρ on Solution Quality:**
 - **Scenario 1: ($\alpha=1.0$, $\beta=2.0$, $\rho=0.5$):** By placing higher importance on the physical heuristic ($\beta=2.0$), the ants heavily favored physically shorter links over historical trails. This allowed the colony to instantly identify the direct edge (0 -> 4), matching Dijkstra's exact route.
 - **Scenario 2: ($\alpha=2.5$, $\beta=1.0$, $\rho=0.3$):** By placing heavy emphasis on pheromone trails ($\alpha=2.5$), the ants were mathematically biased to follow the earliest successful routes. While it still found an optimal latency of 6, it chose a multi-hop path (0 -> 1 -> 2 -> 4). The low evaporation rate ($\rho=0.3$) meant the colony quickly locked into this multi-hop route and heavily reinforced it, demonstrating how high alpha values can cause rapid convergence on the first "good enough" path discovered by the swarm.